Introduction to Anatomy Ontologies

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Outline

- General anatomy concepts
- Why ontologies
- What is an ontology
- Hierarchies
- Knowledge Graphs
- Anatomy ontologies
  - Standardizations
- Interoperability

Anatomy

- Facts based on direct observation
- What you see

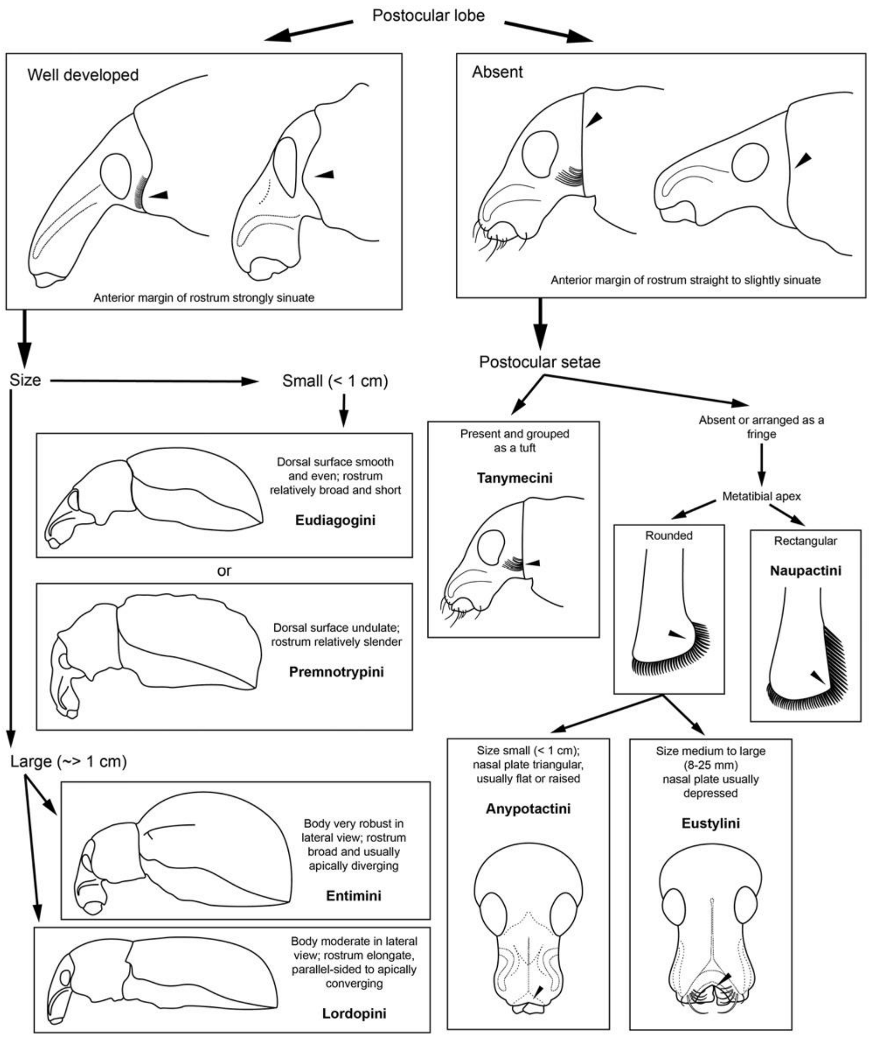
Morphology

- Hypotheses to explain the facts of anatomy
  - Functional
  - Evolutionary
- What you interpret from what you see

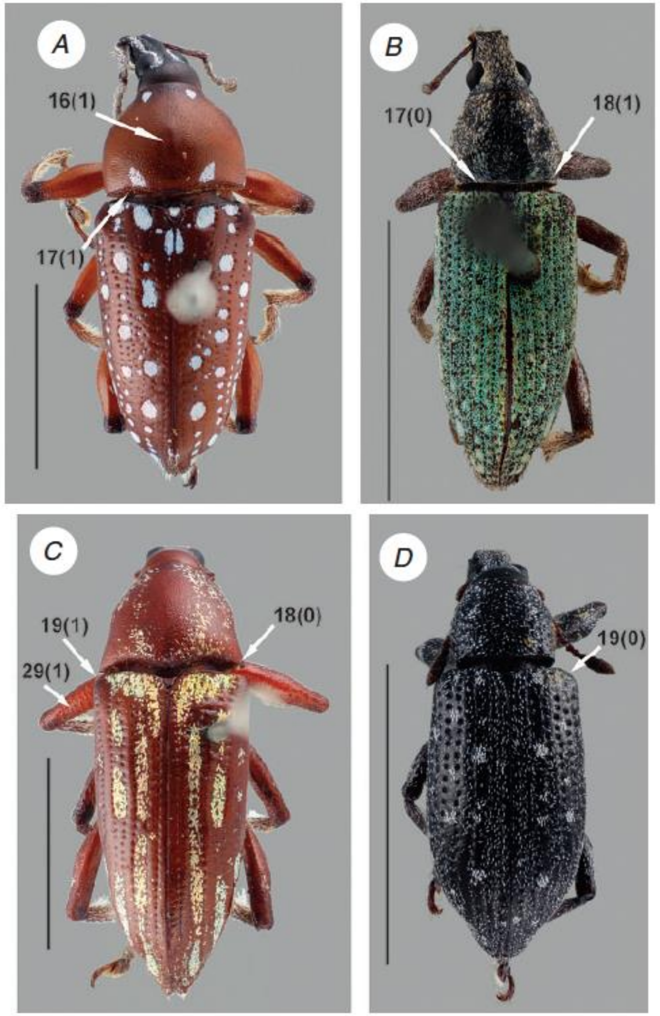
Snodgrass, R.E. (1951) Anatomy and Morphology. *Journal of the New York Entomological Society* 59, 71–73.
https://www.jstor.org/stable/25005398

Anatomical structures in Systematics

Diagnostics



Character coding



Descriptions

Genus *Katasophistes* Girón & Short, 2018
Figs 2, 5, 39, 40A–D

Katasophistes Girón & Short, 2018: 132.

Gender. Masculine.

Type species. *Katasophistes merida* Girón & Short, 2018: 136; by original designation.

Diagnosis. Medium to small beetles, body length 2.7–4.5 mm. Body shape oval to elongated in dorsal view; moderately and evenly convex in lateral view (Fig. 39). Color orange brown to dark brown, rather uniform along body regions (Fig. 39). Shape of head trapezoid. Eyes relatively small, subquadrate, at most only slightly emarginated anteriorly, moderately projected from outline of head. Clypeus trapezoid, with anterior margin broadly emarginate. Labrum fully exposed. Mentum with strong median anterior depression sometimes limited by low transverse carina; surface of mentum with lateral oblique ridges (Fig. 39C, F). Antennae with nine antennomeres; cupule slightly asymmetric, with rounded outline. Maxillary palps moderately long, 0.7 × to nearly as long as width of head; inner margin of maxillary palpomere 2 slightly curved near apex, outer margin curved, sometimes strongly, along apical half (Fig. 39C, F). Each elytron with five rows of deep/large systematic punctures; elytra without sutural striae, with outer margins slightly flared; serial punctures absent (Fig. 39A, D). Prosternum slightly convex to tectiform. Posterior elevation of mesoventrite, with a well-defined, curved transverse ridge; anapleural sutures forming an obtuse angle, separated at anterior margin by distance 0.2–0.3 × the width of anterior margin of mesepisternum. Metaventrite densely pubescent, except for large median rhomboid glabrous patch (Fig. 39C, F). Protibiae with spines of anterior row hair-

Natural language

- How humans communicate with each other

An ontology is a tool to
formally organize knowledge

Formal (logic) language

- Understandable by humans AND processable by machines
- Representation of terms, definitions, and relations (obo/owl)
- Simplest form – controlled vocabularies

The OBO Foundry

- Principles – Open science
- **Interoperability**
  - BFO – Basic Formal Ontology
  - UBERON – multi-species anatomy ontology
  - BSPO – Biological Spatial Ontology
  - PATO – Phenotype And Trait Ontology
  - RO – Relation Ontology

The Ontology Development Kit (ODK)

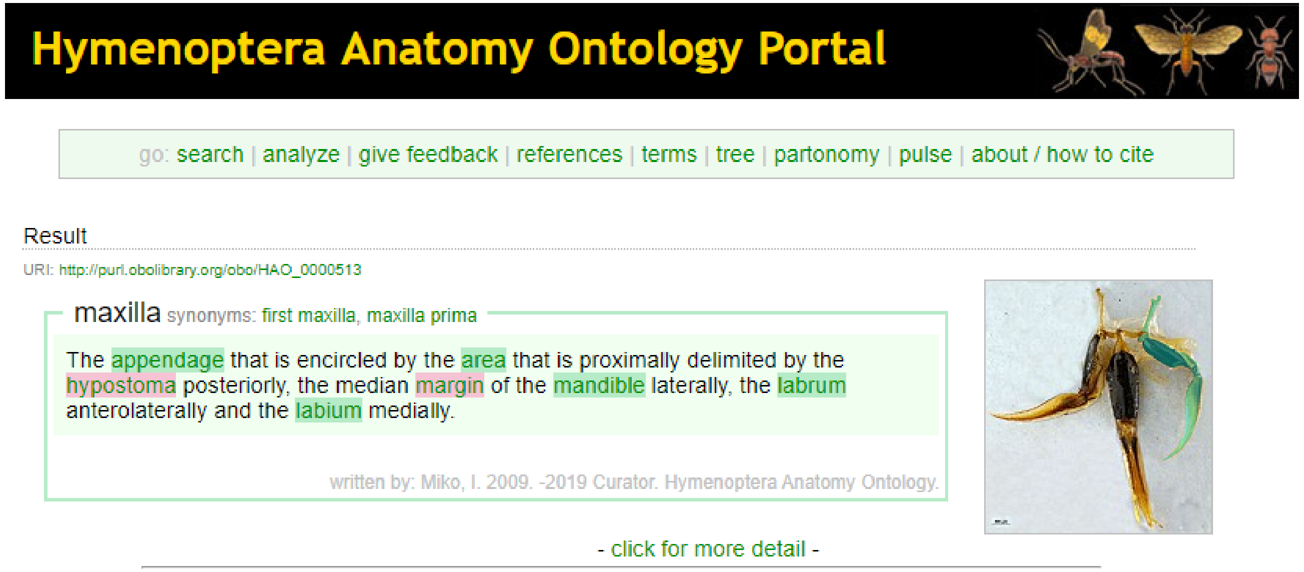
Build the ODK images and run the tests | passing | DOI | 10.5281/zenodo.4973944

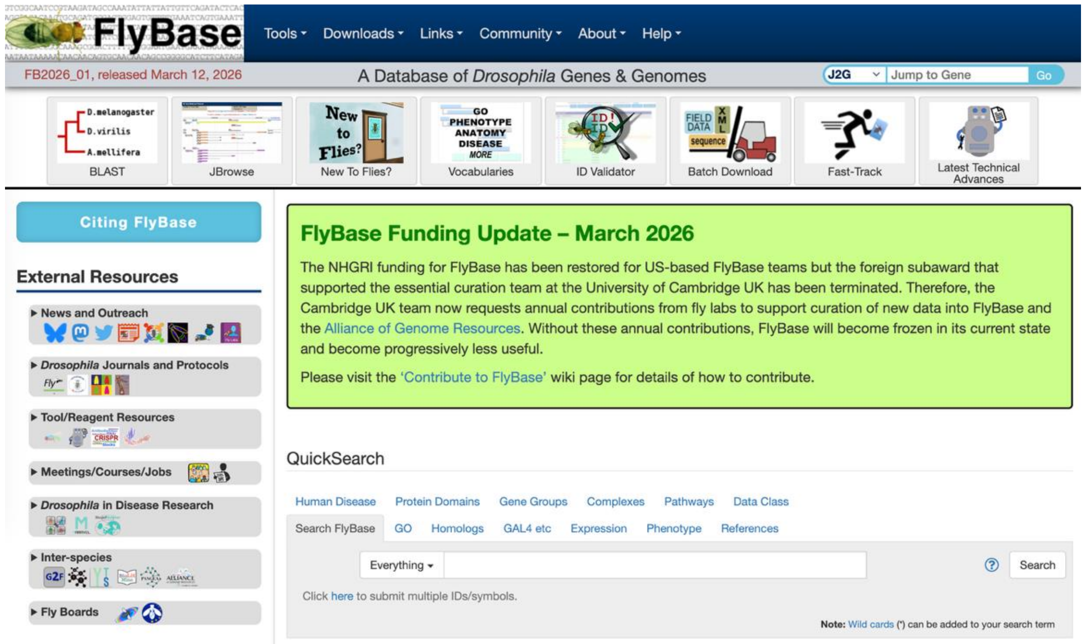
Initialize a Git repo for managing your ontology the OBO Library way!

GitHub

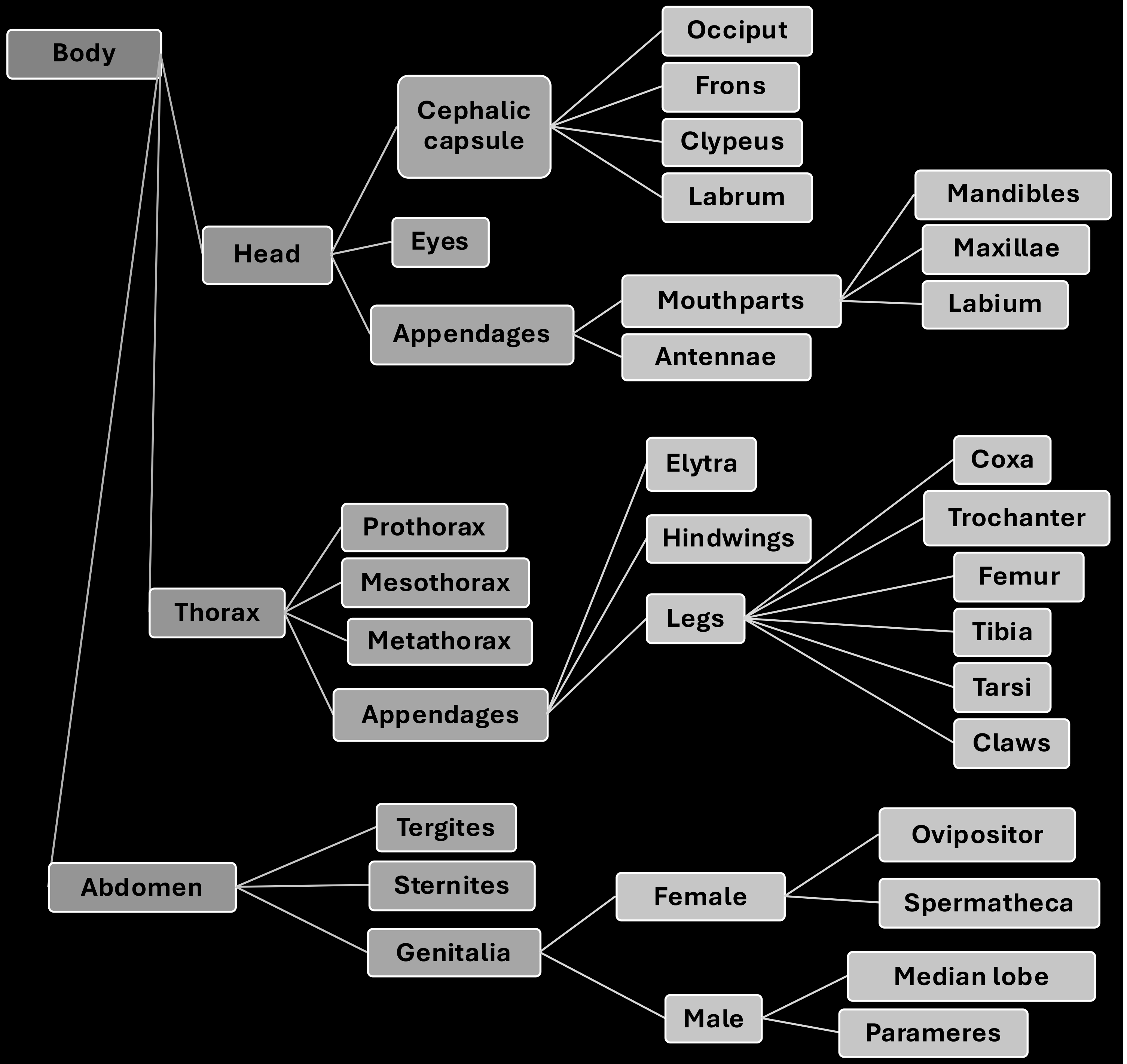
Allows collaborative editing

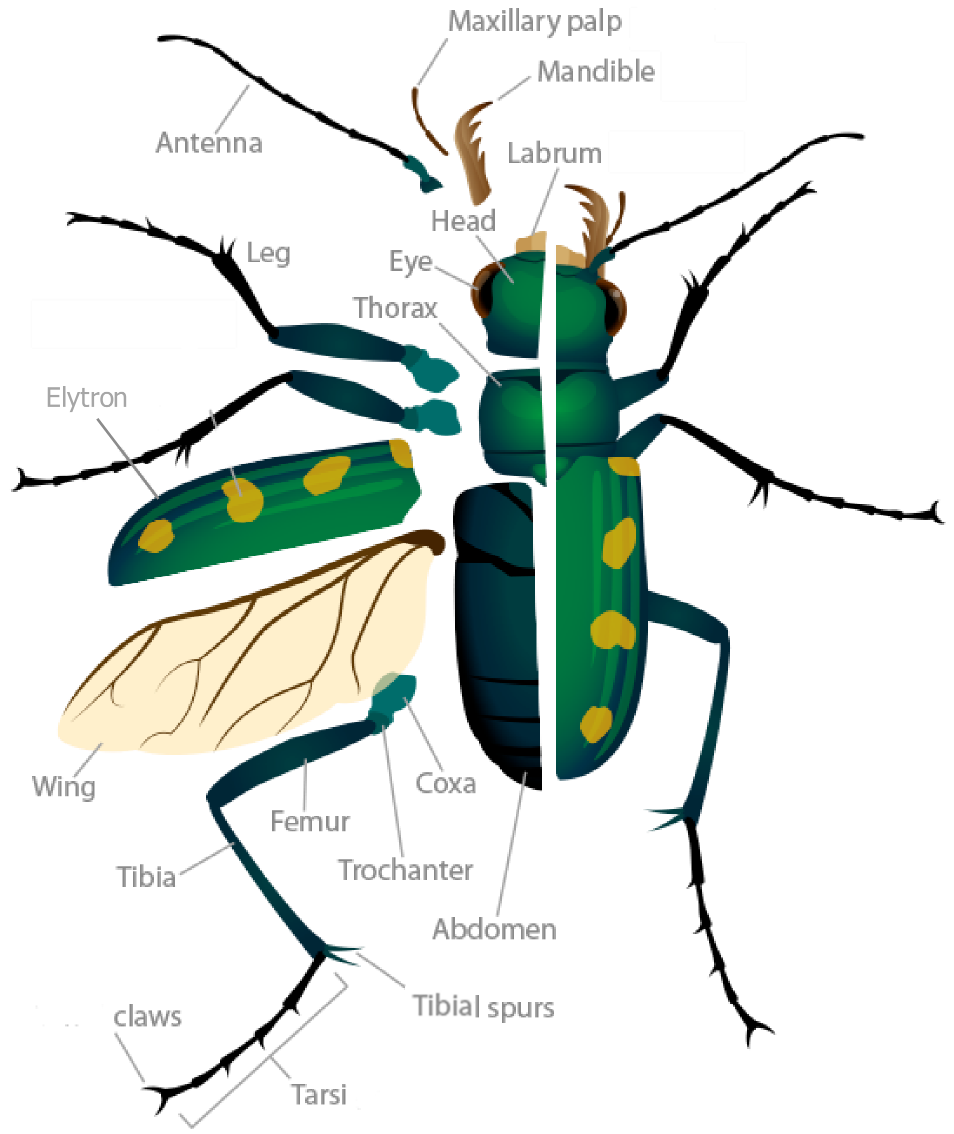
Anatomy Ontologies for Insects





- **Ontology for the Anatomy of the Insect SkeletoMuscular system (AISM)**
- **Coleoptera Anatomy Ontology (COLAO)**
- Hymenoptera Anatomy Ontology
- *Drosophila* anatomy ontology
- *Drosophila* Phenotype Ontology
- Mosquito Gross Anatomy Ontology
- Lepidoptera Anatomy Ontology
- *Tribolium* Ontology
- Collembola Anatomy Ontology
- Spider Anatomy Ontology
- Tick Anatomy Ontology
- Ontology of Arthropod Circulatory Systems


Body
Head
Cephalic capsule
Occiput
Frons
Clypeus
Labrum
Eyes
Appendages
Mouthparts
Mandibles
Maxillae
Labium
Antennae
Thorax
Prothorax
Mesothorax
Metathorax
Appendages
Elytra
Hindwings
Legs
Coxa
Trochanter
Femur
Tibia
Tarsi
Claws
Abdomen
Tergites
Sternites
Genitalia
Female
Ovipositor
Spermatheca
Male
Median lobe
Parameres

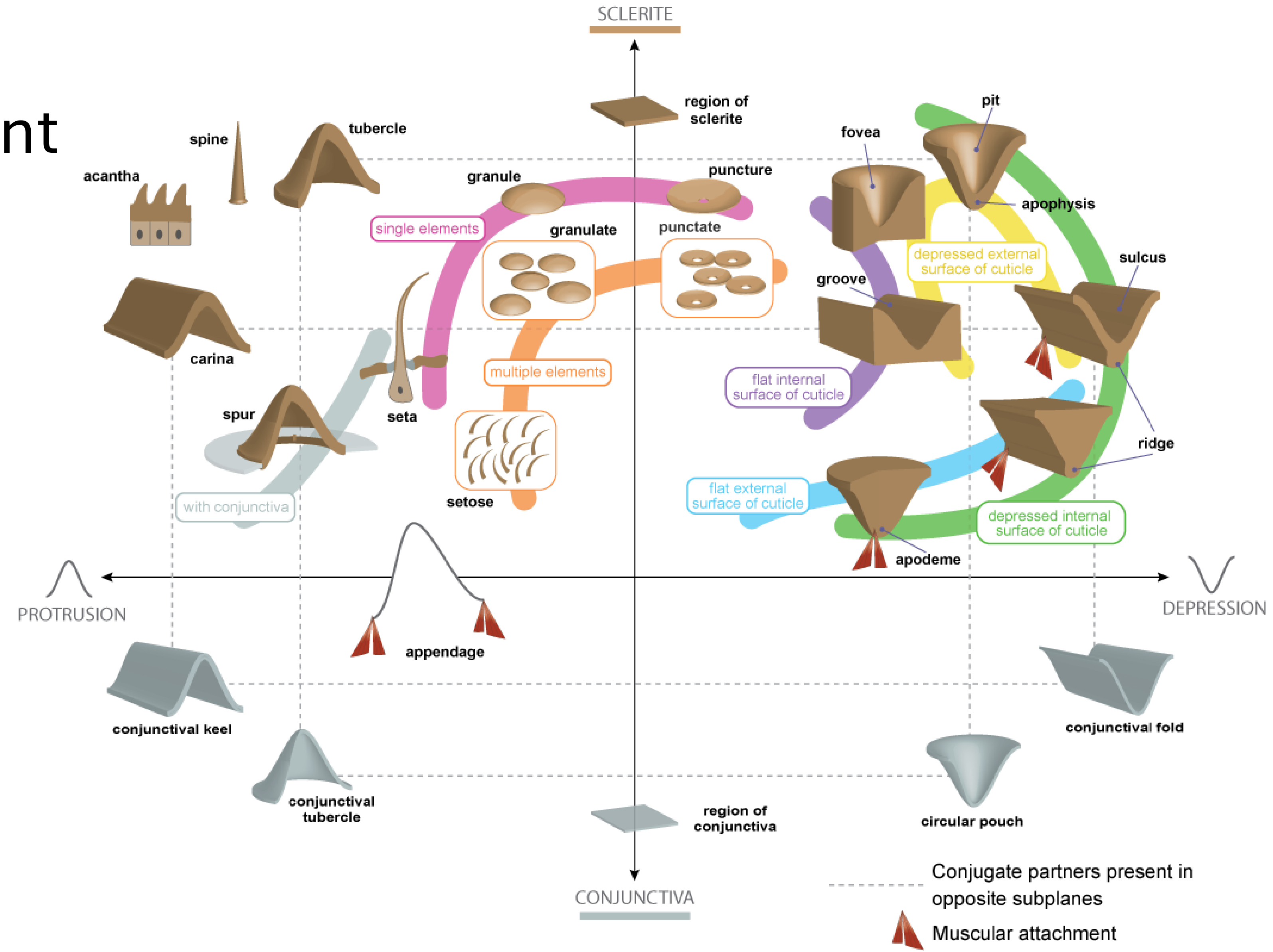

Maxillary palp
Mandible
Antenna
Labrum
Head
Leg
Eye
Thorax
Elytron
Wing
Coxa
Femur
Trochanter
Tibia
Abdomen
Tibial spurs
claws
Tarsi

Cuticular system - Structural properties

Kind of cuticular element

- Sclerite
  - Laminar
  - Ring-like
- Conjunctiva

- Features
  - Protrusion
  - Depression



Girón et al. 2023

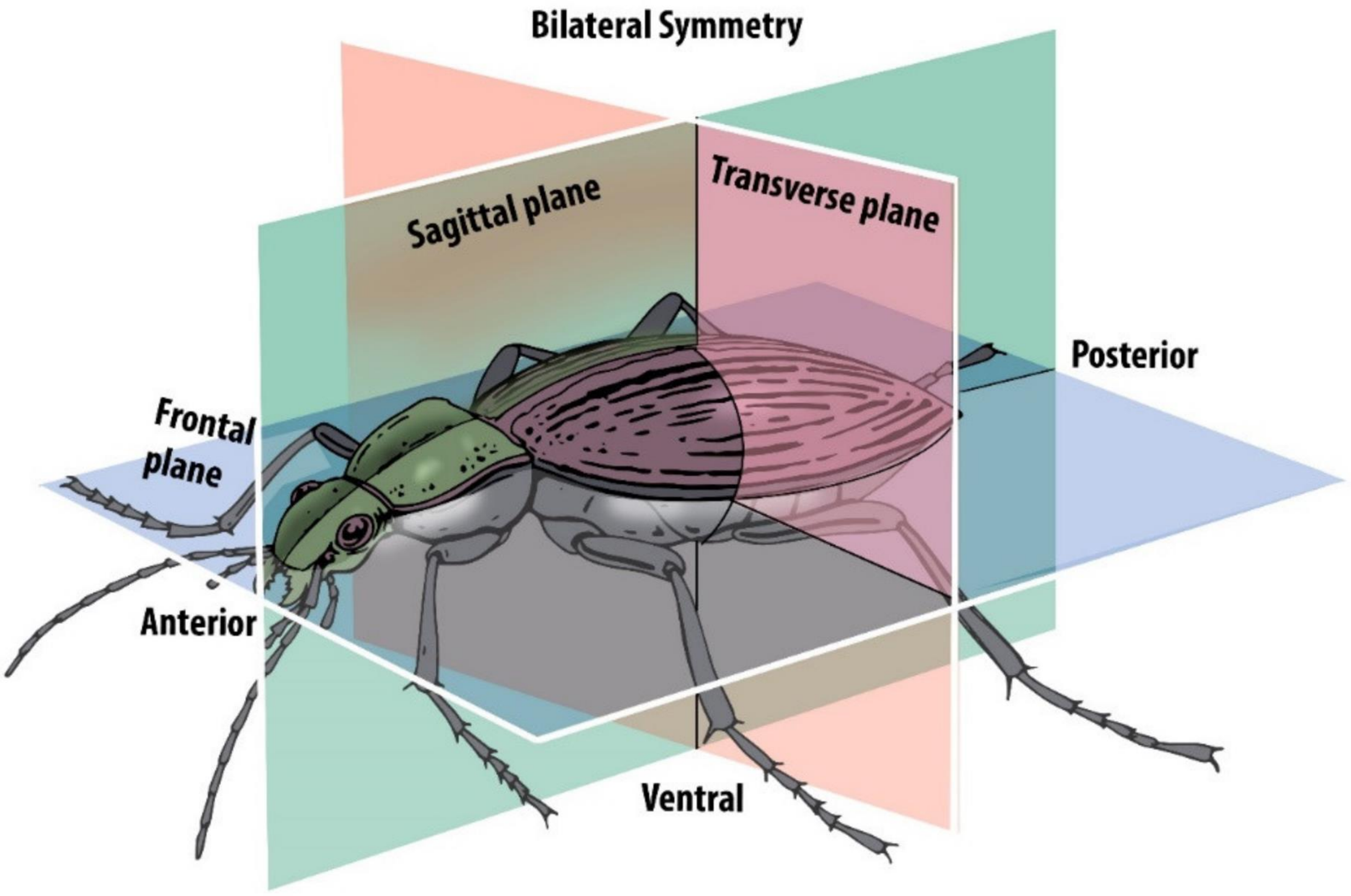
Cuticular system – Orientation – Positional relationships

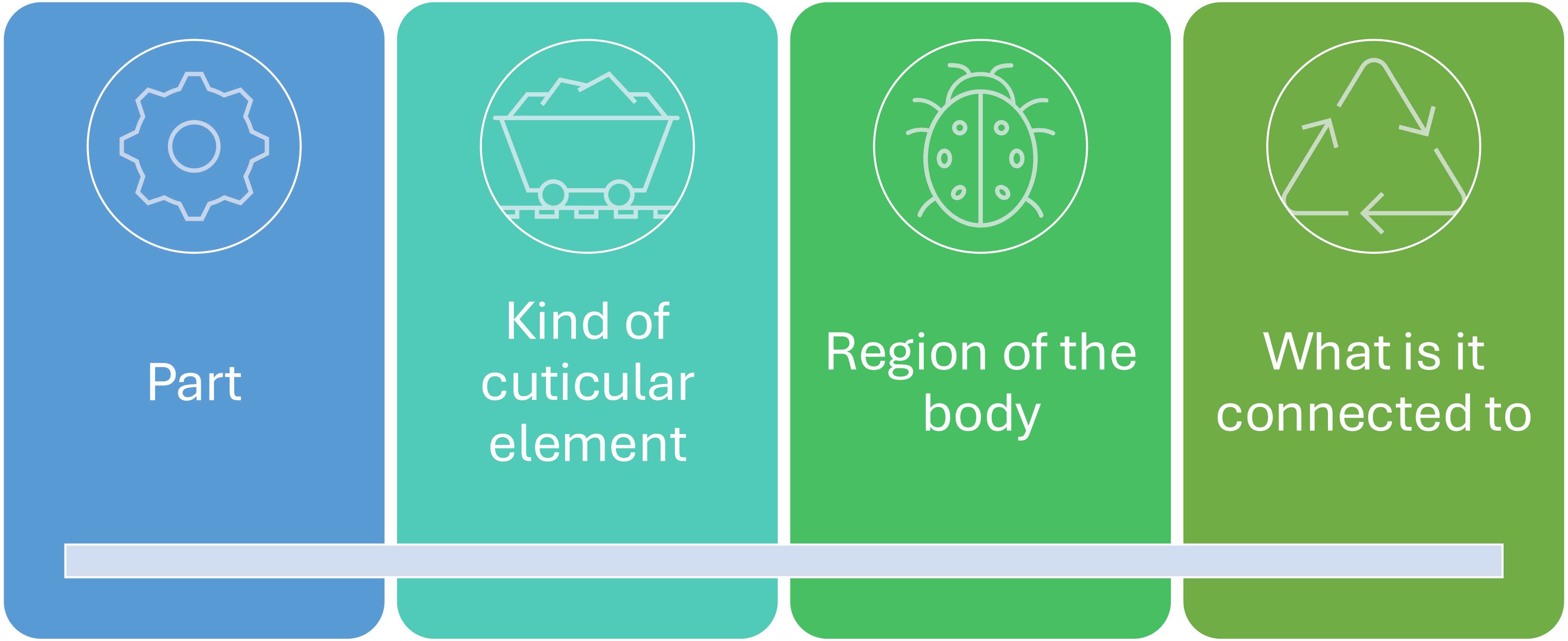
Locators

- body tagma
- body region
  - Dorsal
  - Ventral

  - Anterior
  - Posterior

  - Lateral
  - Mesal





Multiple aspects

Each aspect with multiple categories

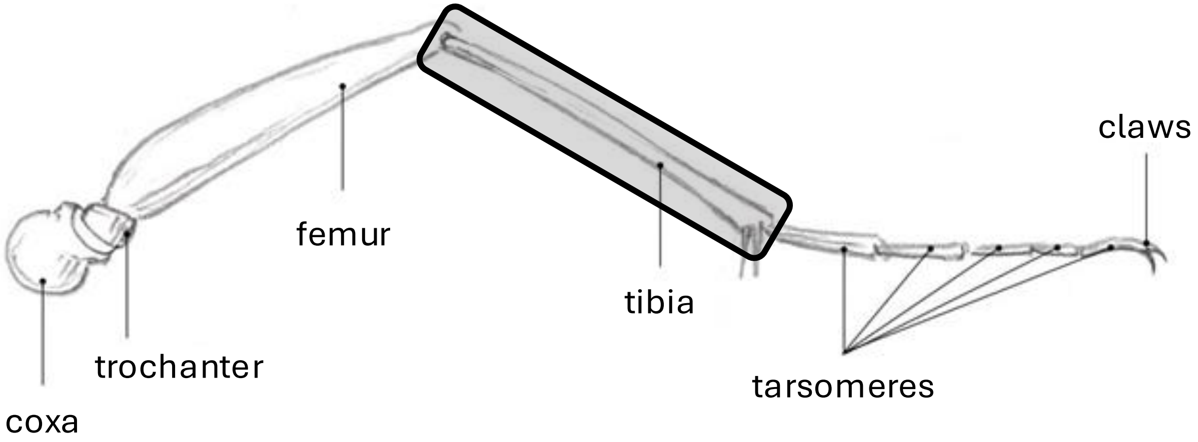
Categories can be organized hierarchically

ontology

ontology: representation of terms, definitions, and relations

Natural language

Logical language



https://johnmuirlaws.com/draw-insects-understanding-drawing-legs/

Object	Property	Subject
tibia	part of	leg
leg	has part	tibia

Part

Kind of cuticular element

Region of the body

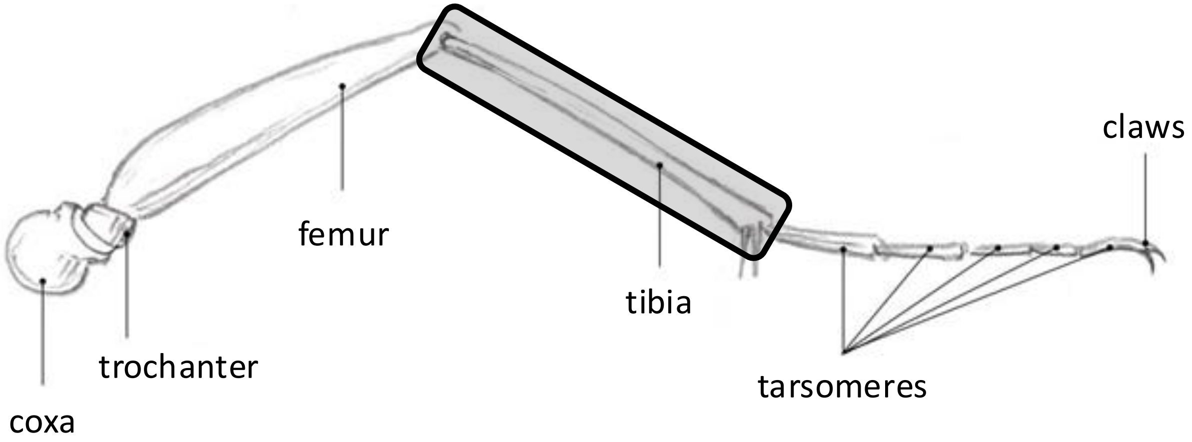
What is it connected to

Anatomy ontologies: systems of multiple hierarchies

tibia

part of

leg



https://johnmuirlaws.com/draw-insects-understanding-drawing-legs/

Part

Kind of cuticular element

Region of the body

What is it connected to

Body

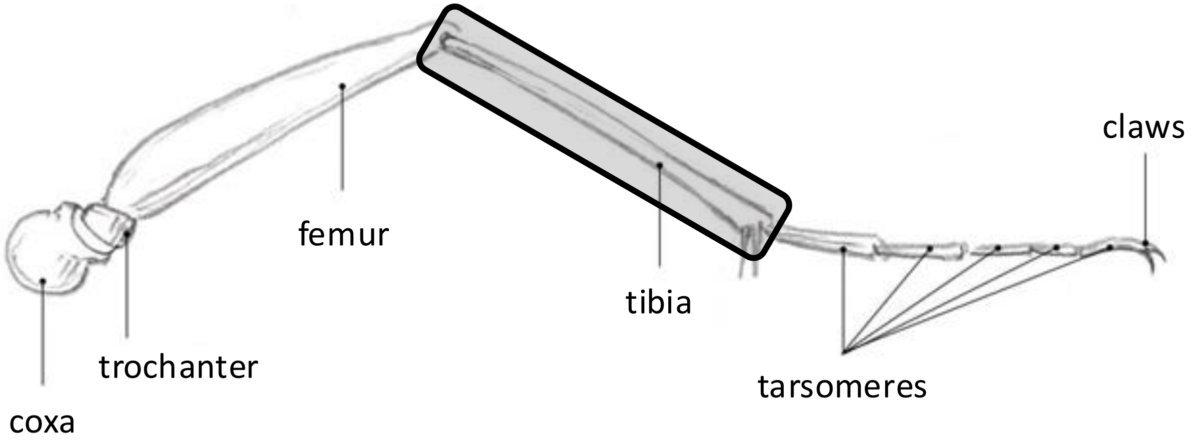
Cuticle

Region

Connected to

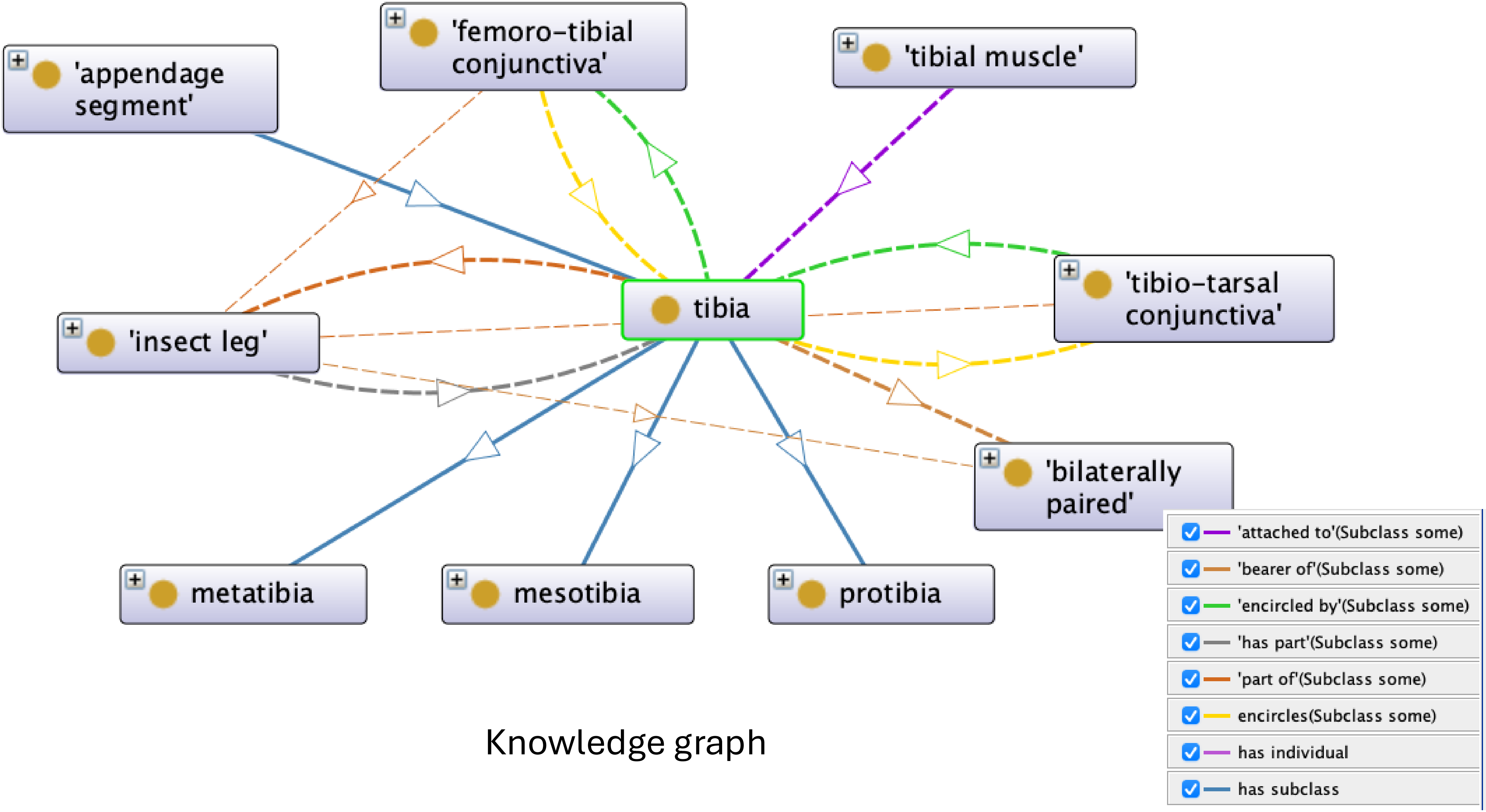
Anatomy ontologies: systems of multiple hierarchies

properties are logically associated (inherited)



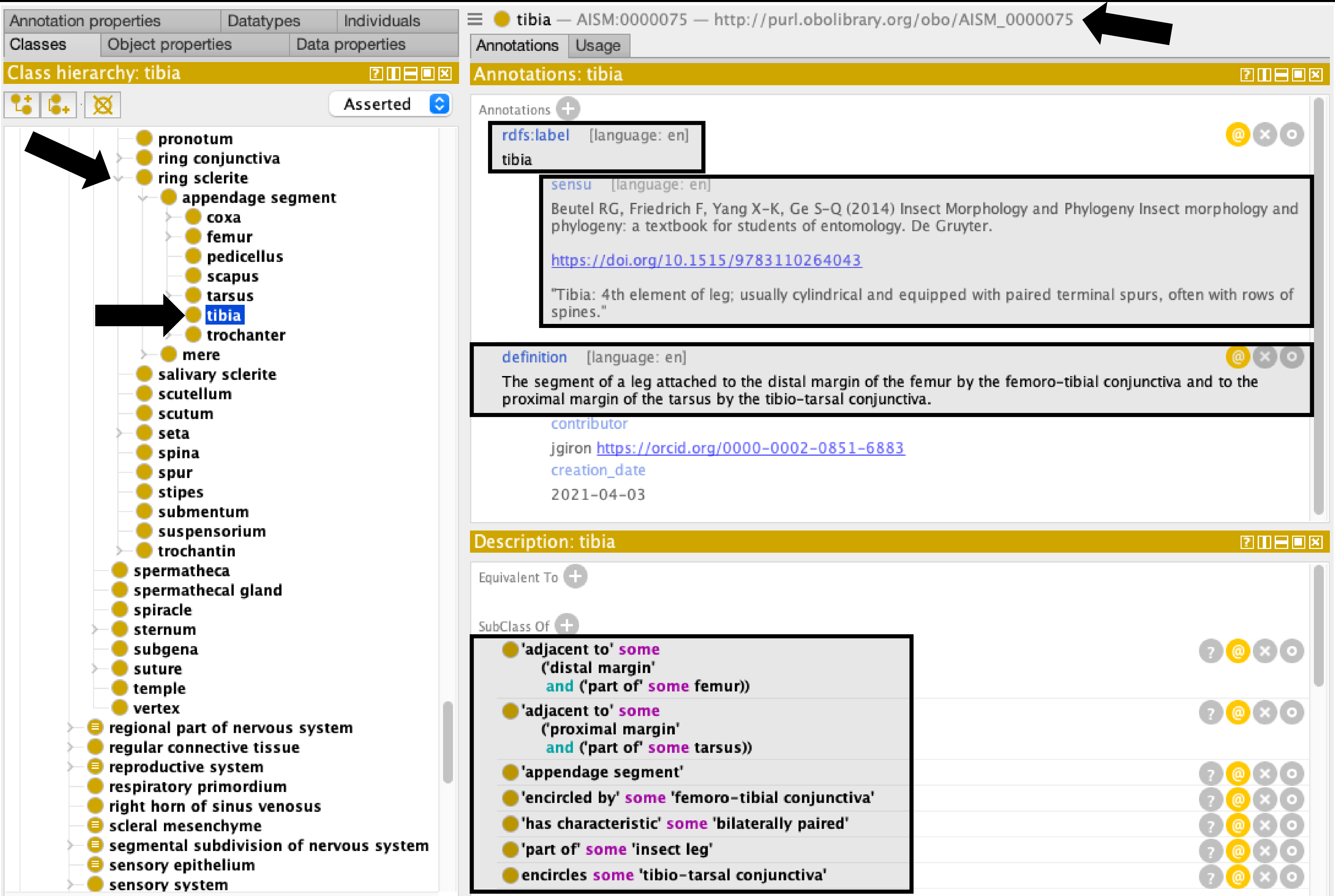
https://johnmuirlaws.com/draw-insects-understanding-drawing-legs/

- part of the insect leg
  - part of insect thorax
- appendage segment
  - ring sclerite
  - bilaterally paired
- adjacent to the distal margin of the femur
  - encircled by the femoro-tibial conjunctiva
- adjacent to the proximal margin of the tarsus
  - encircles the tibio-tarsal conjunctiva



Knowledge graph

Tibia in Protégé



Terms and definitions must be unique within ontology

Important for reasoning

Definitions

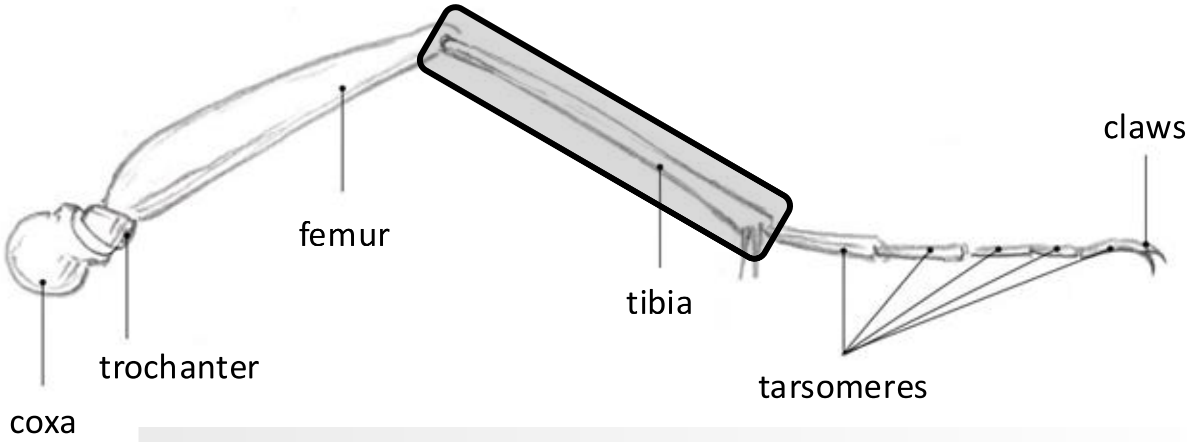
Logical

Natural

Contain the same elements

Interoperability

UBERON - multi-species anatomy ontology



tibia

http://purl.obolibrary.org/obo/AISM_0000075 Copy

Definition:

The cuticle of the segment of a leg attached to the distal margin of the femur by the femoro-tibial conjunctiva.

tibia

http://purl.obolibrary.org/obo/UBERON_0000979 Copy

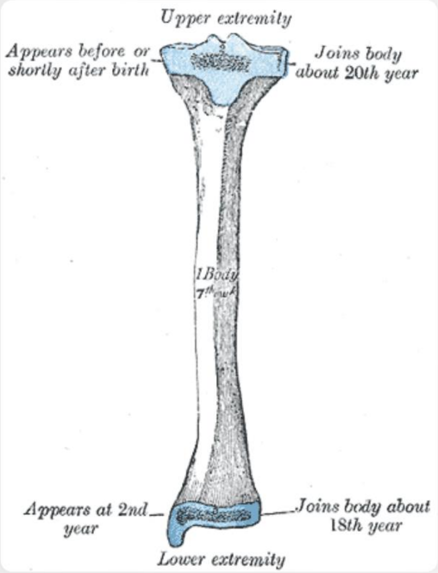
Definition:

The major preaxial endochondral bone in the posterior zeugopod[Phenoscape].

Also appears in CL CPONT DOID EFO FOVT + 10

Exact Synonyms shankbone shinbone

Depicted by



Vertebrate biased

Check that the terms and definitions you are using are appropriate

BSPO - Biological Spatial Ontology

Biological Spatial Ontology

Version 2023-05-27

An ontology for respresenting spatial concepts, anatomical axes, gradients, regions, planes, sides and surfaces. These concepts can be used at multiple biological scales and in a diversity of taxa, including plants, animals and fungi. The BSPO is used to provide a source of anatomical location descriptors for logically defining anatomical entity classes in anatomy ontologies.

Imports from BFO CARO DC DCTERMS GO *+ 8*

Exports to AISM CL CLO COLAO COVOC *+ 34*

proximal

Search

Jump to

proximal side	BSPO	BSPO:0000061
proximal to	BSPO	BSPO:0000100
proximal region	BSPO	BSPO:0000077
proximal surface	BSPO	BSPO:0000377
proximal margin	BSPO	BSPO:0000677

Search OLS for proximal

Interoperability

PATO - Phenotype And Trait Ontology

- quality
  - color
  - shape
  - size
  - and many other attributes
    - chemical
    - physiological
    - behavioral

- BFO - Basic Formal Ontology

Class Hierarchy

Thing
- + BFO_0000020
  - + quality
    - + qualitative
    - + process quality
    - + variability
    - + increased quality
    - + decreased quality
    - + dendrite quality
    - + quantitative
    - - physical object quality
      - + morphology
      - + physical quality
      - + functionality
      - + disposition
      - + organismal quality
      - + population quality
      - + molecular quality
      - + aplastic/hypoplastic
      - + decreased object quality
      - + increased object quality
      - - structurally discontinuous
      - + absence of physical object
      - more...

Interoperability

RO - Relation Ontology

- has_part- BFO:0000051
- part_of- BFO:0000050
- has_characteristic- RO:0000053
- characteristic_of- RO:0000052
- increased_in_magnitude_relative_to- RO:0015007
- decreased_in_magnitude_relative_to- RO:0015008

phenoscript

Interoperability

- GO - Gene Ontology
- CL - Cell Ontology
- ChEBI - Chemical Entities of Biological Interest
- NCBI taxonomy ontology

Where to search for ontologies or terms?

https://obofoundry.org/

OBO Foundry About Principles Ontologies Citation Participate Newsletter FAQ

Open Biological and Biomedical Ontology Foundry

Community development of interoperable ontologies for the biological sciences

Learn about OBO best practices and community resources

- OBO Foundry principles
- OBO tutorial
- Ontology browsers, tutorials, and tools

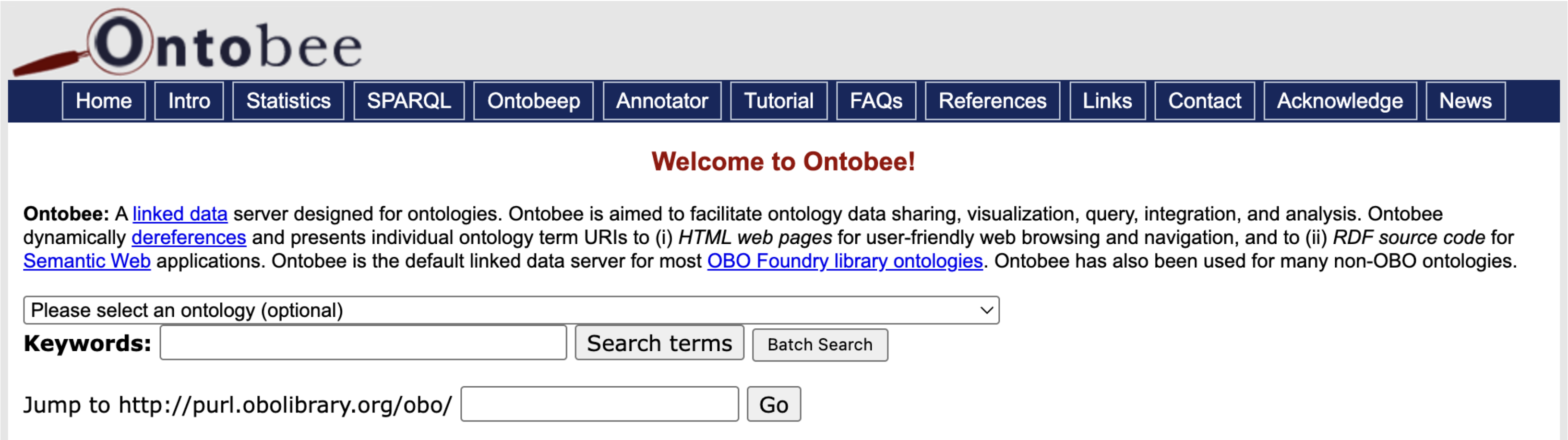
Participate

- Code of Conduct
- Join the OBO mailing list and the OBO Community Slack workspace
- OBO Foundry Operations and Working Groups
- Submit bug reports or suggestions for improvement via GitHub
- Submit your ontology to be considered for inclusion in the OBO Foundry

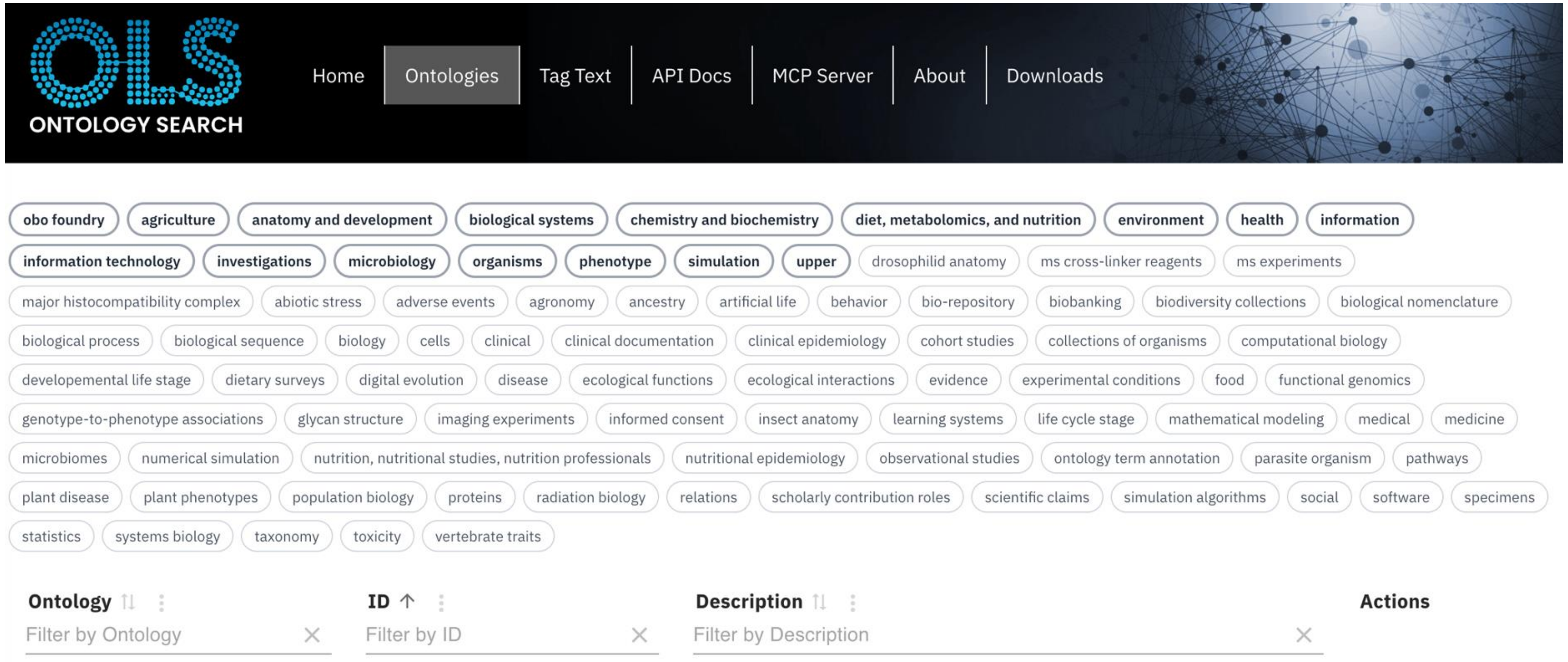
Find, use, and contribute to community ontologies

Download table as: [YAML | JSON-LD | RDF/Turtle]

Where to search for ontologies or terms?



https://ontobee.org/



https://www.ebi.ac.uk/ols4/

Questions?

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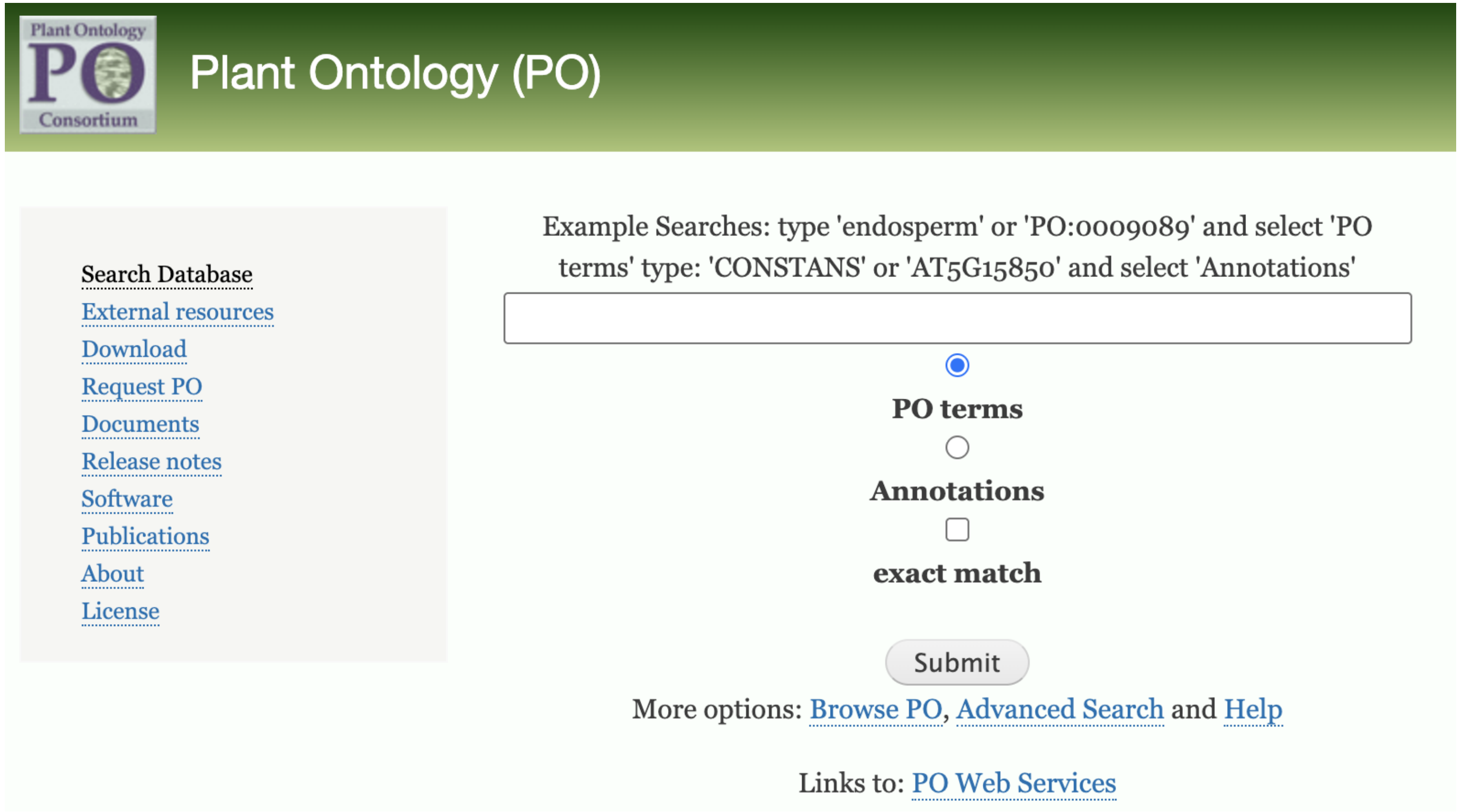
Plant ontologies



- Plant Ontology (PO)
  - Plant anatomy
  - Structure and development
- Plant Traits (TO)
- Plant Phenology Ontology (PPO)

Vertebrate ontologies

Uberon Multi-species Anatomy Ontology

Uberon Ontology Documentation

Quick links:

- Uberon on Github
- Issue Tracker
- Browse Uberon on OLS
- About Uberon
- The Uberon Team

- UBERON
- FuTRES Ontology of Vertebrate Traits (FOVT)
- Vertebrate Taxonomy Ontology (VTO)